

Full Weil positivity and the surviving mixed-support trace

Exact prime energies and the original companion boundary

Programme continuation with explicit source provenance

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This continuation receives the actual degree- p operator on the relative principal parts and computes its action on the original mixed-support tests. The resulting signed joint observation admits both signs even on compact convolution squares satisfying both Weil moments. We then evaluate the full Archimedean term and prove that the entire three-pulse family has strictly positive complete Weil form, with the exact lower bound $5981N/52500$. These tests are therefore excluded as full-Weil counterexamples. Their mixed support state remains nonzero; after retaining the common Archimedean term, its joint trace is strictly negative at every tensor order at least two. The complete maps and proofs of these assertions follow.

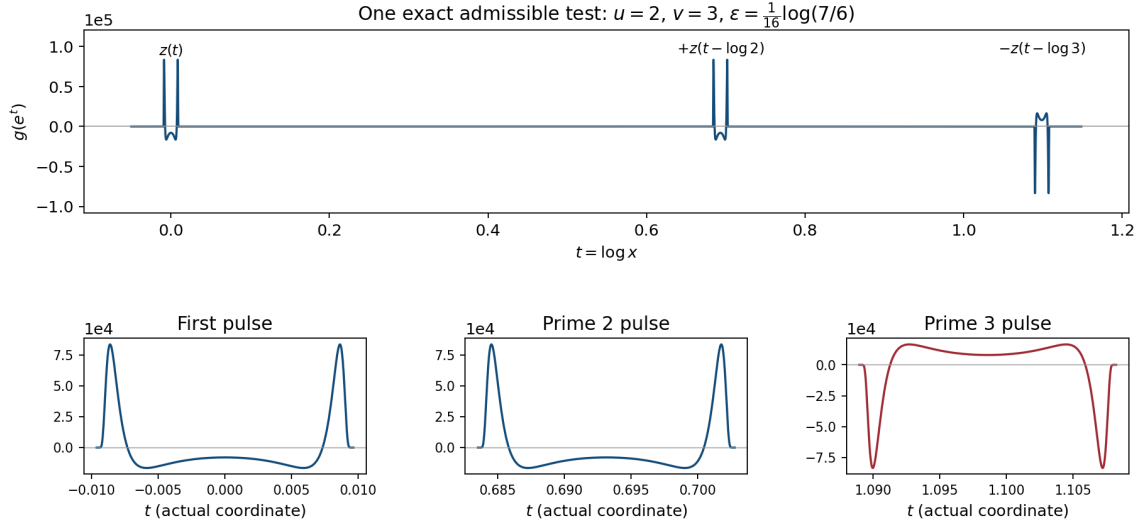
The companion calculation constructs the original off-critical q -dimensional object as a recurrence subspace in the actual local tensor operator's annulus distribution channel. It retains every original cardinal label, phase, source metric, support label and observation-kernel component. Evaluation at zero is its exact inverse. Its exponential growth and its two finite boundary terms are calculated without identifying that distribution subspace with the global interior cohomology. No RH decision is claimed.

Human sources and programme inputs. The analytic convention is Alain Connes's original author TeX, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, section `sectweilpos`, equations `wp` and `arch`, which attributes the positivity criterion to André Weil. The source of the relative principal parts and local degree operator is the supplied 031 proof, RP21–38, PD1–21 and IA1–9. The full-lattice carrier, original companion coordinates and mixed support diagonal are the supplied 032 proof, CLM8–17, CR1–41, PDD1–15, JMD1–23 and TDP1–4. Their complete, exact-version TeX sources accompany this paper in the reproducibility archive, under `inputs/031_SOURCE.tex` and `inputs/032_SOURCE.tex`; the source account records their hashes. These supplied witnesses are identified as such, with no unverified public-edition claim. The full local operator proofs and all input definitions needed for the new assertions are included again where used below.

The supported-zero prime chain is prior foundational work. The user-provided origin attribution is Gemini 2.5 Pro; the first [globalization source, prime-ideal classification and dimension corollary](#), has an empty author field, while the later full-lattice Version 11 has the byline The Clankers. These identities are not merged. The general purity comparison belongs to Pierre Deligne, *La conjecture de Weil. II*, especially 3.3.6 for the image of compactly supported cohomology in ordinary cohomology. No application of that theorem to the constructed annulus space is assumed here. The proofs below do not require an unverified transcription of Deligne's theorem. Earlier ES–Fable provenance remains with its original Levent Alpöge credit in the retained input collection; no new ES result is asserted in this continuation.

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$$\int g(e^t) e^{t/2} dt = \int g(e^t) e^{-t/2} dt = 0$$

$$W_2(f) = \alpha_2 > 0, \quad W_3(f) = -\alpha_3 < 0, \quad W_p(f) = 0 \quad (p \neq 2, 3)$$

$$Q(g) > N/10 > 0$$

$$\text{Tr}[D_{\mathbf{h}} Q(f)'] < 0 \quad (r \geq 2)$$

Exact test and moments: PME21–25. Full Weil lower bound: FWP. Full four-state trace: FWJ1–8. N is the unaltered integral of z^2 ; $\alpha p = 2N(\log p)/\sqrt{p}$. The plots evaluate the stated formulas.

Figure 1: The actual three-pulse test for primes 2 and 3, in its original logarithmic coordinate and amplitude. PME21–25 proves both Mellin conditions and every prime evaluation; FWP1–24 proves the complete Weil lower bound with Connes’s full Archimedean term. FWJ1–8 proves the simultaneously negative signed support trace, including that same Archimedean term. The expanded pulse panels show the indicated coordinate intervals without changing the pulse amplitudes.

1 The relative prime degree on admissible mixed-support tensors

The calculation in this section composes the actual prime-degree operator of the 031 continuation with the compact tests and the support diagonal of the 032 continuation. It establishes both signs for their mixed observation on genuine admissible convolution squares. The complete Weil sign for the constructed three-pulse family is calculated in Section 2.

Sources, authorship, and scope. The prior inputs are the programme's 031 working source `01_REBUILT_MATHEMATICS.tex`, equations PD1–PD21, RP36–RP38, and IA1–IA9, and the 032 working source, equations PDD1–PDD15 and JMD1–JMD23. The new computation is the exact composition proved below. The finite-place distributions and Mellin convention are Alain Connes's original author TeX, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, section `sectweilpos`, equation `wp`; that section attributes the positivity criterion to André Weil. The supported-zero foundation is an earlier programme result, not a discovery of this calculation. The user's attribution of its original derivation to Gemini 2.5 Pro is retained separately from the supplied original source's empty author field and the later Version 11 byline The Clankers. Its exact original source is `globalization_note.tex`, supported-zero prime theorem. No class-semigroup claim is used here.

1.1 Independent verification of the prime-degree receiver

Fix a rational prime p . Retain $|p|_p = p^{-1}$ and additive Haar measure $\mu(\mathbb{Z}_p) = 1$. Put $D_p = \mathbb{Z}[1/p]/\mathbb{Z} = \mathbb{Q}_p/\mathbb{Z}_p$. The equality is the map induced by rational inclusion: its kernel is zero because $a/p^n \in \mathbb{Z}_p$ implies $p^n \mid a$, and its surjectivity follows by taking the finite negative part of a p -adic expansion. Multiplication by p on D_p is onto and has exactly p elements in every fibre; its kernel consists of the classes a/p . On counting-measure $\ell^2(D_p)$ the actual pullback and its adjoint are

$$(A_p f)(d) = f(pd), \quad (A_p^* f)(u) = \sum_{pd=u} f(d), \quad A_p^* A_p = pI. \quad (\text{PME1})$$

These formulas first hold on finite-support functions. Fibre counting gives $\|A_p f\|^2 = p\|f\|^2$, so they extend boundedly and $V_p = p^{-1/2} A_p$ is an isometry. Its factor is fixed by the degree.

For $d = a/p^n$ define $\chi_d(x) = \exp(2\pi i a(x \bmod p^n)/p^n)$ on $\mathbb{Z}_p$. Changing representatives or increasing n leaves the character unchanged. The finite geometric sum proves orthogonality of these characters for Haar measure one. At each level they are a basis of the functions on $\mathbb{Z}/p^n\mathbb{Z}$, and those functions are dense in $L^2(\mathbb{Z}_p)$. Thus $\delta_d \mapsto \chi_d$ is unitary. Under this Fourier convention,

$$V_p F(x) = \sqrt{p} \mathbf{1}_{p\mathbb{Z}_p}(x) F(x/p), \quad V_p^* F(x) = p^{-1/2} F(px). \quad (\text{PME2})$$

Indeed the sum of $\chi_d(x)$ over $pd = u$ vanishes unless $x \in p\mathbb{Z}_p$, and for $x = py$ it is $p\chi_u(y)$. Changing variables $x = py$, $dx = p^{-1}dy$, proves the adjoint. The range projection of V_p^n is multiplication by $\mathbf{1}_{p^n\mathbb{Z}_p}$, whose decreasing ranges have zero intersection because the limiting singleton has Haar measure zero.

On $\mathcal{H}_p = L^2(\mathbb{Q}_p)$ define

$$(\mathcal{U}_p F)(x) = \sqrt{p} F(x/p), \quad (\mathcal{U}_p^{-1} F)(x) = p^{-1/2} F(px), \quad \xi_p = \mathbf{1}_{\mathbb{Z}_p}, \quad r_p = p^{-1/2}. \quad (\text{PME3})$$

The change of variables above proves unitarity. Extension by zero embeds the space in PME2 and intertwines its isometry with $\mathcal{U}_p$. The spaces supported in $p^{-n}\mathbb{Z}_p$ exhaust $\mathcal{H}_p$, so this is the minimal unitary extension. For $n \geq 0$, $\mathcal{U}_p^n \xi_p = p^{n/2} \mathbf{1}_{p^n\mathbb{Z}_p}$. Integration, and then unitarity for negative n , give

$$\langle \mathcal{U}_p^n \xi_p, \xi_p \rangle = r_p^{|n|}, \quad d\nu_p(\theta) = \frac{1}{2\pi} \frac{1 - r_p^2}{1 - 2r_p \cos \theta + r_p^2} d\theta. \quad (\text{PME4})$$

The measure assertion follows by summing the two absolutely convergent geometric series and then expanding the squared norm of every Laurent polynomial applied to ξ_p . These polynomial pairings determine the cyclic measure by density of trigonometric polynomials.

For precision the actual weight-one operator is $T_p^{\text{rel}} = \sqrt{p}\mathcal{U}_p$, not the isometry V_p . It satisfies

$$(T_p^{\text{rel}})^* T_p^{\text{rel}} = T_p^{\text{rel}} (T_p^{\text{rel}})^* = pI, \quad \sigma(T_p^{\text{rel}}) = \{z : |z| = \sqrt{p}\}. \quad (\text{PME5})$$

To prove the entire spectrum assertion, the annuli $p^n \mathbb{Z}_p^\times$, $n \in \mathbb{Z}$, partition $\mathbb{Q}_p$ outside a null singleton. Their L^2 spaces are the mutually orthogonal translates of $L^2(\mathbb{Z}_p^\times)$ by $\mathcal{U}_p^n$. On this direct sum $\mathcal{U}_p$ is the bilateral shift. For a unit vector b in the zeroth summand and $|\lambda| = 1$, the unit vectors $(2N+1)^{-1/2} \sum_{n=-N}^N \lambda^{-n} \mathcal{U}_p^n b$ have squared $(\mathcal{U}_p - \lambda)$ -error $2/(2N+1)$. For $|\lambda| \neq 1$ a Neumann series for $\mathcal{U}_p$ or its inverse gives the inverse of $\mathcal{U}_p - \lambda$. This proves PME5 after scaling. An eigenvector would have equally large components in all annular summands, so square summability forces every component to vanish. There are no nonzero eigenvectors. These checks validate PD1–PD13 and PD20–PD21 with their stated measures, Fourier signs, and distinctions between the operators.

1.2 The exact bounded energy map and its signed operator

Let $H = L^2(\mathbb{R}_+^\times, dx/x)$, with inner product linear in the first variable. For $g \in C_c^\infty(\mathbb{R}_+^\times)$ put $F(t) = g(e^t)$, $a_p = \log p$, and retain the actual map

$$(\mathcal{E}_p g)(x) = \sum_{n \in \mathbb{Z}} F(x + na_p) \mathcal{U}_p^n \xi_p, \quad 0 \leq x < a_p, \quad \mathcal{E}_p : H \longrightarrow K_p := L^2([0, a_p]; \mathcal{H}_p). \quad (\text{PME6})$$

The sum is finite on the initial domain. Write $g^*(x) = \overline{g(x^{-1})}$ and $f = g * g^*$, using Haar convolution dx/x . Expanding the norm and applying PME4, then setting $j = n - m$ and $t = x + na_p$, proves

$$\begin{aligned} \|\mathcal{E}_p g\|^2 &= \sum_{j \in \mathbb{Z}} r_p^{|j|} \int_{\mathbb{R}} F(t) \overline{F(t - ja_p)} dt \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{g}(s)|^2 P_{r_p}(a_p s) ds, \quad P_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2}, \quad \hat{g}(s) = \int_0^\infty g(x) x^{-is} \frac{dx}{x}. \end{aligned} \quad (\text{PME7})$$

For fixed j the intervals in the change of variables partition $\mathbb{R}$ up to endpoints. Fourier inversion gives the second line. Interchange with the Poisson series is justified by its absolute sum $(1+r_p)/(1-r_p)$ times the integrable function $|\hat{g}(s)|^2$. Consequently PME6 extends uniquely boundedly, with $\|\mathcal{E}_p\|^2 = (1+r_p)/(1-r_p)$: the upper bound follows from PME7 and is attained as a supremum by functions whose Fourier mass is supported near $s = 0$, where P_{r_p} attains its maximum. Polarization of PME7 identifies the bounded self-adjoint operator

$$B_p = (\log p)(\mathcal{E}_p^* \mathcal{E}_p - I_H) \quad \text{with multiplier} \quad b_p(s) = (\log p)(P_{r_p}(a_p s) - 1). \quad (\text{PME8})$$

Its exact range, hence its essential range and spectrum, is

$$\sigma(B_p) = \left[-\frac{2r_p \log p}{1 + r_p}, \frac{2r_p \log p}{1 - r_p} \right]. \quad (\text{PME9})$$

The denominator of P_r ranges continuously from $(1-r)^2$ to $(1+r)^2$. Every neighborhood of each attained value has a preimage of positive Lebesgue measure. Multiplication by $b_p - z$ has bounded inverse outside the closed range, while supported unit vectors near any value in the range are approximate eigenvectors. This proves PME9 without transferring an unproved spectral assertion.

The $j = 0$ summand of PME7 is $\|g\|^2$. Hence the entire prime-power distribution in Connes's stated convention is

$$W_p(f) = (\log p) \sum_{m \geq 1} p^{-m/2} [f(p^m) + f(p^{-m})] = \langle B_p g, g \rangle = (\log p)(\|\mathcal{E}_p g\|^2 - \|g\|^2). \quad (\text{PME10})$$

This verifies PD14–PD17 and their signs. It also agrees with RP36–RP38: the trace-class diagonal operator $D_p(s)e_m = p^{-ms}e_m$, $m \geq 1$, has trace $\sum_{m \geq 1} p^{-ms}$ for $\Re s > 0$. Substituting $s = 1/2 \pm it$ into both traces and using Fourier inversion gives exactly PME10, including $1/(2\pi)$ and every $\log p$. The sum of absolute trace coefficients is $2r_p/(1-r_p)$, which justifies the integral interchange. Similarly the scalar resolvent on ξ_p is $(1-zr_p)^{-1}$ for $|z| < 1$; its scalar series extends to $|z| < \sqrt{p}$ without an operator resolvent assertion there. This verifies PD19's stated distinction.

The calculation exhibits why PME5 alone does not make PME10 nonnegative: the exact receiving map subtracts I_H , and its operator has both signs by PME9. No local degree or measure factor is changed in this statement.

1.3 Composition with the original PDD tests

Retain every PDD coefficient. For each prime p , set

$$\begin{aligned} \varepsilon_p &= \frac{1}{8} \log((p+1)/p), & u_p(t) &= \begin{cases} e^{-1/(1-(t/\varepsilon_p)^2)}, & |t| < \varepsilon_p, \\ 0, & |t| \geq \varepsilon_p, \end{cases} \\ v_p &= u_p'' - \frac{1}{4}u_p, & N_p &= \int_{\mathbb{R}} |v_p(t)|^2 dt. \end{aligned} \quad (\text{PME11})$$

Every derivative of the bump tends to zero at the endpoints, because the exponential dominates each inverse power of $1 - (t/\varepsilon_p)^2$. Thus the displayed functions are smooth. Integration by parts gives $\int v_p u_p = -\int |u_p'|^2 - \frac{1}{4} \int u_p^2 < 0$, so $N_p > 0$. For $\sigma \in \{1, -1\}$ put

$$g_{p,\sigma}(e^t) = v_p(t) + \sigma v_p(t - a_p), \quad f_{p,\sigma} = g_{p,\sigma} * g_{p,\sigma}^*. \quad (\text{PME12})$$

Two integrations by parts, including their zero boundary terms, give $\int v_p(t)e^{ct}dt = (c^2 - 1/4) \int u_p(t)e^{ct}dt = 0$ for $c = 1/2$ and $c = -1/2$. Translation multiplies this zero by e^{ca_p} . Therefore both original Mellin values of $g_{p,\sigma}$ at $\pm i/2$ vanish.

Let $C_p(b) = \int v_p(t)v_p(t-b)dt$. It is even, supported in $[-2\varepsilon_p, 2\varepsilon_p]$, and $C_p(0) = N_p$. Expanding the convolution gives $f_{p,\sigma}(e^b) = 2C_p(b) + \sigma C_p(b - a_p) + \sigma C_p(b + a_p)$. For every integer $n \geq 2$, $n \neq p$, $|\log n - a_p| \geq \log((p+1)/p) = 8\varepsilon_p$; below p use $p/(p-1) > (p+1)/p$, and above p use monotonicity. Also $2\varepsilon_p < \log 2$ and $2\varepsilon_p < a_p$. These facts prove all the evaluations, and their energy receivers,

$$\begin{aligned} \|g_{p,\sigma}\|^2 &= 2N_p, \\ W_q(f_{p,\sigma}) &= 2\sigma(\log p)r_p N_p \mathbf{1}_{q=p}, \\ \|\mathcal{E}_q g_{p,\sigma}\|^2 &= 2N_p(1 + \sigma r_p \mathbf{1}_{q=p}). \end{aligned} \quad (\text{PME13})$$

Here unique factorization ensures $q^m = p$ only for $q = p, m = 1$. The last line follows by the proved PME10; it computes the actual energy map for every prime, not only the chosen one.

There is also a direct vector calculation at $q = p$:

$$(\mathcal{E}_p g_{p,\sigma})(x) = \begin{cases} v_p(x)(\xi_p + \sigma \mathcal{U}_p \xi_p), & 0 \leq x < \varepsilon_p, \\ v_p(x - a_p)(\mathcal{U}_p^{-1} \xi_p + \sigma \xi_p), & a_p - \varepsilon_p < x < a_p, \\ 0, & \varepsilon_p \leq x \leq a_p - \varepsilon_p. \end{cases} \quad (\text{PME14})$$

Each nonzero case comes from exactly two terms of PME6. Both vector norms squared are $2(1 + \sigma r_p)$ by PME4, and the two integrals of v_p^2 add to N_p . This independently verifies its PME13 value.

1.4 The two-place support trace and all four signed energies

Fix distinct rational primes u, v . In the four-state space $V = \mathbb{C}[\mathcal{P}(\{u, v\})]$, retain

$$\mathbf{h} = \delta_{\emptyset} - \delta_{\{u\}} - \delta_{\{v\}} + \delta_{\{u,v\}}, \quad \mathcal{V}_r \delta_A = \left(\sum_{p \in A} W_p \right)^{\otimes r}. \quad (\text{PME15})$$

This is the original linearized support diagonal followed by the local distribution maps, not the tensor power of $\mathcal{V}_1 \mathbf{h}$. Every tensor factor is a separate multiplicative variable. Expanding the four support states gives

$$\mathcal{V}_2 \mathbf{h} = W_u \otimes W_v + W_v \otimes W_u. \quad (\text{PME16})$$

On $H \otimes H$ its exact bounded receiving operator is $\mathbb{B}_{u,v} = B_u \otimes B_v + B_v \otimes B_u$. For $G = g_1 \otimes g_2$ and $f_j = g_j * g_j^*$, PME10 proves $\langle \mathbb{B}_{u,v} G, G \rangle = \mathcal{V}_2 \mathbf{h}(f_1 \otimes f_2)$. The full energy identity, valid for every $G \in H \otimes H$, is

$$\begin{aligned} \langle \mathbb{B}_{u,v} G, G \rangle &= (\log u)(\log v) [\\ &\quad \|(\mathcal{E}_u \otimes \mathcal{E}_v)G\|^2 - \|(\mathcal{E}_u \otimes I)G\|^2 \\ &\quad - \|(I \otimes \mathcal{E}_v)G\|^2 + \|G\|^2 \\ &\quad + \|(\mathcal{E}_v \otimes \mathcal{E}_u)G\|^2 - \|(\mathcal{E}_v \otimes I)G\|^2 \\ &\quad - \|(I \otimes \mathcal{E}_u)G\|^2 + \|G\|^2]. \end{aligned} \quad (\text{PME17})$$

Indeed expand PME8 in both tensor factors. Boundedness of all maps permits this identity first on finite tensors and then by continuity.

For the requested literal tests, set $G_{\sigma,\tau} = g_{u,\sigma} \otimes g_{v,\tau}$. Their convolution square on $(\mathbb{R}_+^\times)^2$ is exactly $f_{u,\sigma} \otimes f_{v,\tau}$ by Fubini; all integrals have compact support. In each coordinate its two Mellin conditions vanish, even with the other coordinate left unevaluated. PME13 now gives

$$\boxed{\mathcal{V}_2 \mathbf{h}(f_{u,\sigma} \otimes f_{v,\tau}) = 4\sigma\tau(\log u)(\log v)(uv)^{-1/2}N_u N_v.} \quad (\text{PME18})$$

In PME17 the first four squared norms, after factoring $4N_u N_v$, are respectively $(1 + \sigma r_u)(1 + \tau r_v)$, $1 + \sigma r_u$, $1 + \tau r_v$, and 1. Their signed sum is $\sigma\tau r_u r_v$. The other four squared norms all equal $4N_u N_v$ by the two off-prime cases of PME13 and hence cancel. This proves the displayed value directly from the actual relative-prime energy maps, including every subtracted diagonal term.

The value also has a literal trace realization. Write Π_A for projection onto $\mathbb{C}\delta_A$, $D_{\mathbf{h}}$ for the diagonal matrix with entries $(1, -1, -1, 1)$, and $B_A = \sum_{p \in A} B_p$. Define

$$\mathbb{O}_2 = \sum_A \Pi_A \otimes B_A \otimes B_A, \quad R_G(z) = \langle z, G \rangle G. \quad (\text{PME19})$$

Then $D_{\mathbf{h}} \otimes R_G$ is trace class of rank at most four, $\mathbb{O}_2$ is bounded, and summing their four diagonal blocks gives

$$\text{Tr}_{V \otimes H \otimes H} [\mathbb{O}_2 (D_{\mathbf{h}} \otimes R_G)] = \langle \mathbb{B}_{u,v} G, G \rangle. \quad (\text{PME20})$$

For each block, the elementary rank-one trace identity is $\text{Tr}(BR_G) = \langle BG, G \rangle$, as follows by expansion in an orthonormal basis. Thus PME20 is exactly JMD14 with the local distribution insertions received by PME10. It is not the trace of the identity on the invariant line $\mathbb{C}\mathbf{h}$, and it does not identify these insertions with the programme's complete analytic cohomological trace.

1.5 A single admissible test with both local signs

The negativity can also be achieved with the same test in every tensor factor. This removes the decoding-function limitation of JMD22 while preserving its original mixed support state. Put $M = uv$ and specify the new width and bump literally by

$$\begin{aligned} \varepsilon_{u,v} &= \frac{1}{16} \log((M+1)/M), \quad w(t) = \begin{cases} e^{-1/(1-(t/\varepsilon_{u,v})^2)}, & |t| < \varepsilon_{u,v}, \\ 0, & |t| \geq \varepsilon_{u,v}, \end{cases} \\ z &= w'' - \frac{1}{4}w, \quad N = \int_{\mathbb{R}} z(t)^2 dt > 0. \end{aligned} \quad (\text{PME21})$$

The smoothness and strict positivity follow by exactly the endpoint argument and the integration by parts displayed after PME11, now with this stated width. No original PDD function is rescaled or replaced; PME21 defines an additional test. For $\sigma, \tau \in \{1, -1\}$ set

$$g_{u,v;\sigma,\tau}(e^t) = z(t) + \sigma z(t - \log u) + \tau z(t - \log v), \quad f_{u,v;\sigma,\tau} = g_{u,v;\sigma,\tau} * g_{u,v;\sigma,\tau}^*. \quad (\text{PME22})$$

All three terms have each exponential moment at $c = \pm 1/2$ equal to zero, by the same explicit integration by parts. Therefore $\hat{g}_{u,v;\sigma,\tau}(i/2) = \hat{g}_{u,v;\sigma,\tau}(-i/2) = 0$.

Let $C(b) = \int z(t)z(t-b)dt$, $a = \log u$, and $b_0 = \log v$. The complete correlation, including the new cross-place lags, is

$$f_{u,v;\sigma,\tau}(e^s) = 3C(s) + \sigma[C(s-a) + C(s+a)] + \tau[C(s-b_0) + C(s+b_0)] + \sigma\tau[C(s-a+b_0) + C(s+a-b_0)]. \quad (\text{PME23})$$

Here C is even, supported in $[-2\varepsilon_{u,v}, 2\varepsilon_{u,v}]$, and $C(0) = N$. We check every extra lag rather than discarding it. For an integer $n \geq 2$, $n \neq u$, the gap $|\log n - \log u|$ is at least $\log((u+1)/u)$; the corresponding statement holds at v . Both gaps exceed $16\varepsilon_{u,v}$. The arguments $\log n + \log u$, $\log n + \log v$, and $\log n$ are positive and at least $\log 2 > 16\varepsilon_{u,v}$. The remaining two arguments are $\log(nv/u)$ and $\log(nu/v)$. Neither is zero: $nv = u$ or $nu = v$ would contradict distinct primality and $n \geq 2$. For any positive integer $A \neq d$,

$$|\log(A/d)| \geq \log((d+1)/d) \quad (d \geq 2). \quad (\text{PME24})$$

For $A > d$, use $A \geq d+1$; for $A < d$, use $d/(d-1) > (d+1)/d$. Apply PME24 with $(A, d) = (nv, u)$ or (nu, v) . Their gaps exceed $16\varepsilon_{u,v}$ as well. This proves that the cross-place lags never reach a positive prime power. At $s = 0$ the nonzero differences between the three centers also have absolute value greater than $16\varepsilon_{u,v}$: for $|\log(u/v)|$ use PME24 with the smaller denominator when appropriate, and the same lower bound with M . Consequently the complete evaluations are

$$\begin{aligned} \|g_{u,v;\sigma,\tau}\|^2 &= 3N, \\ W_q(f_{u,v;\sigma,\tau}) &= 2N[(\log u)r_u\sigma\mathbf{1}_{q=u} + (\log v)r_v\tau\mathbf{1}_{q=v}], \\ \|\mathcal{E}_q g_{u,v;\sigma,\tau}\|^2 &= 3N + 2N[r_u\sigma\mathbf{1}_{q=u} + r_v\tau\mathbf{1}_{q=v}]. \end{aligned} \quad (\text{PME25})$$

The reciprocal evaluations agree by evenness of PME23. The energy formula is now PME10, so it receives the same prime degree as before.

Choose $\sigma = 1$, $\tau = -1$ and put $\alpha_u = 2N(\log u)r_u > 0$, $\alpha_v = 2N(\log v)r_v > 0$, $f = f_{u,v;1,-1}$. Expansion of the four support states in PME15, for every $r \geq 1$, gives $(W_u + W_v)^{\otimes r} - W_u^{\otimes r} - W_v^{\otimes r}$. Substituting the exact PME25 values proves

$$\boxed{\mathcal{V}_{2m}\mathbf{h}(f^{\otimes 2m}) = (\alpha_u - \alpha_v)^{2m} - \alpha_u^{2m} - \alpha_v^{2m} < 0 \quad (m \geq 1).} \quad (\text{PME26})$$

Strictness follows from $|\alpha_u - \alpha_v| < \max(\alpha_u, \alpha_v)$. In particular the second-order value is $-8N^2(\log u)(\log v)(uv)^{-1/2}$. Taking $\sigma = \tau = 1$ instead gives $(\alpha_u + \alpha_v)^r - \alpha_u^r - \alpha_v^r > 0$ for $r \geq 2$ by the positive mixed binomial terms. Thus both signs occur on actual compact convolution squares satisfying both moments. On the product group, $f^{\otimes 2m}$ itself is the convolution square of $g_{u,v;1,-1}^{\otimes 2m}$, with both vanishing moments in every coordinate, by Fubini and the proved one-variable moments.

The negative value has the exact root limit

$$\lim_{m \rightarrow \infty} [-\mathcal{V}_{2m}\mathbf{h}(f^{\otimes 2m})]^{1/(2m)} = \max(\alpha_u, \alpha_v) > 0. \quad (\text{PME27})$$

For $A = \max(\alpha_u, \alpha_v)$ and $a = \min(\alpha_u, \alpha_v) > 0$, the quantity inside the root lies between $A^{2m}[1 - (1 - a/A)^{2m}]$ and $2A^{2m}$. The roots of both bracket factors tend to one; taking logarithms proves the lower assertion also when $a = A$. This proves PME27.

1.6 The exact connection with the full Weil form

For completeness the Archimedean distribution in the same original convention is the literal expression

$$W_{\text{arch}}(f) = (\log(4\pi) + \gamma_E)f(1) + \int_0^\infty [f(e^s) + f(e^{-s}) - 2e^{-s/2}f(1)] \frac{e^{-s/2}}{1 - e^{-2s}} ds, \quad (\text{PME28a})$$

where γ_E is Euler's constant. It follows from the source's x -integral by $x = e^s$, preserving the subtraction also outside the support of f . For the explicit correlations above the bracket vanishes to first order at $s = 0$, so the apparent denominator singularity is integrable. For sufficiently large s , only the subtraction remains and decays exponentially. Thus this gives a specified convergent integral for every Archimedean value below. The convention retained from Connes is $Q(g) = -W_{\text{arch}}(g * g^*) - \sum_p W_p(g * g^*)$ on the two-moment test space. For the tests in PME12 and PME22 the exact full expressions are respectively

$$\begin{aligned} Q(g_{p,\sigma}) &= -W_{\text{arch}}(f_{p,\sigma}) - 2\sigma(\log p)r_p N_p, \\ Q(g_{u,v;\sigma,\tau}) &= -W_{\text{arch}}(f_{u,v;\sigma,\tau}) - \sigma\alpha_u - \tau\alpha_v. \end{aligned} \quad (\text{PME28})$$

All other finite places have been proved to vanish, and the two pole moments vanish; the Archimedean term remains exactly the source term. Thus the mixed negativity PME26 is an evaluation of the specified joint support observable, with its exact bounded-operator and trace maps PME17–PME20. It is not an evaluation of Q . Section 2 evaluates its full Archimedean contribution and proves a positive bound for the three-pulse values in PME28. A positive local Hilbert representation and the local circle identity PME5 do not by themselves erase this explicit signed mixed receiver.

For every $r \geq 1$ the trace realization PME19–PME20 extends by the literal definitions

$$\mathbb{O}_r = \sum_A \Pi_A \otimes B_A^{\otimes r}, \quad \text{Tr} [\mathbb{O}_r(D_{\mathbf{h}} \otimes R_{g^{\otimes r}})] = \mathcal{V}_r \mathbf{h}((g * g^*)^{\otimes r}). \quad (\text{PME29})$$

Each summand is bounded and the inserted operator still has rank at most four. Its block trace is $h_A \langle B_A^{\otimes r} g^{\otimes r}, g^{\otimes r} \rangle = h_A \langle B_A g, g \rangle^r$. Summing the four terms proves PME29, including the even-order negative values PME26 without an unbounded or unspecified trace.

The distributional zero-scalar boundary of IA6–IA9 is also retained in its stated category. On $\mathbb{C}^D \oplus \mathbb{C}$ for $D = \mathbb{Q}/\mathbb{Z}$, its exact actions are $R_e(f, c) = (f(0)\mathbf{1}_D, c)$ and $R_\tau(f, c) = (c\mathbf{1}_D, c)$, obtained by evaluating pullback at each supported class and at the absent class. The vector $\mathbf{1}_D$ has infinite counting norm, so $R_e\delta_0$ cannot be placed in the Hilbert space used in PME1. On finite character polynomials its Fourier transform is δ_0 because all Fourier coefficients are one. No bounded Hilbert action of either boundary operator has been inferred in PME17–PME20. Their original distributional maps are unchanged by this calculation.

2 Strict positivity of the full Weil form on the mixed three-pulse tests

The tests of PME21–PME25 have negative joint mixed-support traces for appropriate signs. Their *full* Weil form nevertheless has the following uniform positive lower bound, with their original widths and amplitudes unchanged:

$$Q(g_{u,v;\sigma,\tau}) > \frac{5981}{52500} N > \frac{N}{10} \quad (u \neq v \text{ rational primes}, \quad \sigma, \tau \in \{1, -1\}). \quad (\text{FWP1})$$

Here N is the literal integral in FWP2 below. We give every estimate, including the Archimedean subtraction and the rational constants. The bound was proposed in the root calculation and is

independently proved here. The original local distributions and the two-moment test convention are Alain Connes's original author TeX, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, section `sectweilpos`, equations `wp` and `arch`; the criterion in that source is attributed to André Weil. PME21–PME27 supply the programme's explicit three-pulse construction and mixed-observation formulas; the definitions needed for this bound and their relevant properties are proved again below.

2.1 The exact tests and their admissibility

Fix distinct rational primes u, v , and retain

$$\begin{aligned} \varepsilon &= \frac{1}{16} \log \frac{uv+1}{uv}, & w(t) &= \begin{cases} \exp[-1/(1-(t/\varepsilon)^2)], & |t| < \varepsilon, \\ 0, & |t| \geq \varepsilon, \end{cases} & z &= w'' - \frac{1}{4}w, \\ N &= \int_{\mathbb{R}} z(t)^2 dt, & a &= \log u, \quad b = \log v, \quad d = |a-b|, \\ g(e^t) &= g_{u,v;\sigma,\tau}(e^t) = z(t) + \sigma z(t-a) + \tau z(t-b), & f &= g * g^*. \end{aligned} \tag{FWP2}$$

The convolution uses Haar measure dx/x on $\mathbb{R}_+^\times$, and $g^*(x) = \overline{g(x^{-1})}$. Each derivative of w tends to zero at its two endpoints: the displayed exponential dominates every inverse power of $1 - (t/\varepsilon)^2$ arising under differentiation. Thus w, z, g are smooth and compactly supported. Integration by parts proves $\int zw = -\int |w'|^2 - \frac{1}{4} \int w^2 < 0$, so $N > 0$.

For $c = 1/2$ and for $c = -1/2$, two integrations by parts with these zero endpoint terms give $\int z(t)e^{ct}dt = (c^2 - 1/4) \int w(t)e^{ct}dt = 0$. Translation multiplies the integral by e^{ca} or e^{cb} . Consequently

$$\widehat{g}(i/2) = \widehat{g}(-i/2) = 0, \quad \widehat{g}(s) = \int_0^\infty g(x)x^{-is} \frac{dx}{x}. \tag{FWP3}$$

The convolution pole moments also vanish: for $c = \pm 1/2$, Fubini factors the exponential moment of $f(e^t)$ as the moment of $g(e^t)$ at c times the conjugate moment at $-c$. Both factors were just calculated. All such integrals are finite by compact support.

Write $C(s) = \int z(t)z(t-s)dt$. It is real and even, supported in $[-2\varepsilon, 2\varepsilon]$, satisfies $C(0) = N$, and obeys $|C(s)| \leq N$ by Cauchy–Schwarz and translation invariance. Expanding all nine products in the convolution gives exactly

$$\begin{aligned} f(e^s) &= 3C(s) + \sigma[C(s-a) + C(s+a)] + \tau[C(s-b) + C(s+b)] \\ &\quad + \sigma\tau[C(s-d) + C(s+d)]. \end{aligned} \tag{FWP4}$$

For distinct positive integers u, v , the ratio of the larger to the smaller is at least $(\min(u, v) + 1)/\min(u, v)$. It follows, with the literal FWP2 width, that

$$a > 16\varepsilon, \quad b > 16\varepsilon, \quad d > 16\varepsilon. \tag{FWP5}$$

For a, b use $\log 2 > \log((uv+1)/(uv))$; for d use $\min(u, v) < uv$ and strict decrease of $\log((x+1)/x)$. The three translated supports of z are therefore disjoint, so

$$f(1) = \|g\|_{L^2(dx/x)}^2 = 3N, \quad |f(e^s)| \leq 3N \quad (s \in \mathbb{R}). \tag{FWP6}$$

The last inequality is Cauchy–Schwarz applied to the entire translated function $g(e^t)$, retaining its exact norm.

2.2 Every finite prime term

For an integer $A \geq 1$, $A \neq D$, and $D \geq 2$ one has

$$|\log(A/D)| \geq \log((D+1)/D). \tag{FWP7}$$

Above D use $A \geq D + 1$; below D use $D/(D - 1) > (D + 1)/D$. At an integer $n \geq 2$, the term $C(\log n - a)$ in FWP4 vanishes unless $n = u$, by FWP7; its value in that case is N . The analogous term at b vanishes unless $n = v$. All arguments $\log n$, $\log n + a$, and $\log n + b$ exceed 2ε . Finally the two cross-place arguments can be written as $\log(nu/v)$ and $\log(nv/u)$, in some order. They cannot be zero, since distinct primes cannot satisfy $nu = v$ or $nv = u$ for $n \geq 2$. FWP7 with denominator u or v gives a gap exceeding 16ε , so both terms vanish as well. Evenness treats the reciprocal evaluations. Hence the complete local prime distributions in the original convention are

$$\begin{aligned} W_p(f) &= (\log p) \sum_{m \geq 1} p^{-m/2} [f(p^m) + f(p^{-m})], \\ \sum_p W_p(f) &= 2N \left(\sigma \frac{\log u}{\sqrt{u}} + \tau \frac{\log v}{\sqrt{v}} \right). \end{aligned} \tag{FWP8}$$

Unique factorization was used here only to identify a prime power equal to u or v with its first power at that prime. There are no other nonzero finite-place terms.

The function $(\log x)/\sqrt{x}$ on $x > 0$ has derivative $x^{-3/2}(1 - \frac{1}{2} \log x)$, hence maximum $2/e$ at $x = e^2$. For either sign choice in each term, FWP8 therefore gives

$$\sum_p W_p(f) \leq \frac{8N}{e} < 3N. \tag{FWP9}$$

The strict inequality uses the positive Taylor series $e > 1 + 1 + 1/2 + 1/6 = 8/3$. This bound includes all four choices of σ, τ ; no positive sign was tacitly imposed on the prime sum.

2.3 The complete Archimedean subtraction

Keep $c = \log(4\pi) + \gamma_E$ and introduce the positive kernel

$$K(s) = \frac{e^{-s/2}}{1 - e^{-2s}} \quad (s > 0). \tag{FWP10}$$

The original Archimedean integral in Connes's convention, with $x = e^s$, FWP6, and evenness, is exactly

$$W_{\text{arch}}(f) = 3Nc + 2 \int_0^\infty [f(e^s) - 3Ne^{-s/2}] K(s) ds. \tag{FWP11}$$

At zero the bracket is $O(s)$ and $K(s) = O(s^{-1})$, so the integral is locally convergent. Outside the compact support of $f(e^s)$, the subtracted term remains and is exponentially integrable. It is not omitted in any estimate below.

For every $s > 0$, the elementary exponential inequality gives

$$K(s) \leq \frac{1}{1 - e^{-2s}} = 1 + \frac{1}{e^{2s} - 1} \leq 1 + \frac{1}{2s}, \quad 1 - e^{-s/2} \leq s/2. \tag{FWP12}$$

Thus the contribution near zero satisfies

$$\begin{aligned} 2 \int_0^{2\varepsilon} [f(e^s) - 3Ne^{-s/2}] K(s) ds &\leq 6N \int_0^{2\varepsilon} (1 - e^{-s/2}) K(s) ds \\ &\leq N(6\varepsilon^2 + 3\varepsilon). \end{aligned} \tag{FWP13}$$

Indeed substitute FWP6 and FWP12; the resulting integrand is $3N(s + 1/2)$, whose integral has the displayed value.

The entire negative subtraction beyond this interval is evaluated, not estimated away:

$$-6N \int_{2\varepsilon}^\infty e^{-s/2} K(s) ds = -3N \log \coth \varepsilon. \tag{FWP14}$$

To verify the equality, the substitution $t = e^{-s}$ transforms the positive integral into $\int_0^{e^{-2\varepsilon}} (1 - t^2)^{-1} dt = \frac{1}{2} \log((1 + e^{-2\varepsilon})/(1 - e^{-2\varepsilon})) = \frac{1}{2} \log \coth \varepsilon$.

For $s > 2\varepsilon$, the central term in FWP4 vanishes. All the $C(s + a)$, $C(s + b)$, and $C(s + d)$ terms vanish by FWP5. Consequently the remaining positive-half-axis terms are precisely $\sigma C(s - a)$, $\tau C(s - b)$, and $\sigma\tau C(s - d)$. For each center $t \in \{a, b, d\}$ the corresponding interval $[t - 2\varepsilon, t + 2\varepsilon]$ has length 4ε and left endpoint greater than 14ε . FWP12 gives

$$\int_{t-2\varepsilon}^{t+2\varepsilon} K(s) ds \leq 4\varepsilon + \frac{4\varepsilon}{2(t-2\varepsilon)} \leq 4\varepsilon + \frac{1}{7}. \quad (\text{FWP15})$$

Use $|C| \leq N$ on each of the three summands, independently of its sign. This is the exact decomposition bound; applying the weaker $|f| \leq 3N$ separately on three intervals would add an unnecessary factor of three. The intervals need not be disjoint for the following sum bound:

$$2 \int_{2\varepsilon}^{\infty} f(e^s) K(s) ds \leq 6N(4\varepsilon + 1/7). \quad (\text{FWP16})$$

Combining FWP11, FWP13, FWP14, and FWP16 proves the proposed complete estimate with every coefficient retained:

$$\boxed{W_{\text{arch}}(f) \leq 3N(c - \log \coth \varepsilon) + N(27\varepsilon + 6\varepsilon^2 + 6/7)}. \quad (\text{FWP17})$$

2.4 Exact rational constants

Since $uv \geq 6$, one has $\varepsilon \leq \frac{1}{16} \log(7/6)$. For $t > 0$ the identity $(1 + t)^{-1} = 1 - t + t^2 - t^3/(1 + t)$ gives, by integration from zero to $1/6$,

$$\log(7/6) < \frac{1}{6} - \frac{1}{72} + \frac{1}{648} = \frac{25}{162}, \quad 0 < \varepsilon < \frac{25}{2592} < \frac{1}{100}. \quad (\text{FWP18})$$

Also $\tanh t \leq t$ for $t \geq 0$, since its derivative is at most one and it vanishes at zero. Therefore $\coth \varepsilon \geq 1/\varepsilon > 100$. The lower logarithmic bound needed here has a finite rational proof. For $0 < t < 1$, integrate the finite geometric expansion of $(1 - t^2)^{-1}$ to obtain $\log((1 + t)/(1 - t)) > 2 \sum_{j=0}^m t^{2j+1}/(2j + 1)$. Use $t = 1/3, m = 2$ for $\log 2$ and $t = 1/9, m = 0$ for $\log(5/4)$. Because $100 = 2^6(5/4)^2$, this gives

$$\log 100 > 12 \left(\frac{1}{3} + \frac{1}{81} + \frac{1}{1215} \right) + \frac{4}{9} = \frac{1864}{405} > \frac{23}{5}, \quad \log \coth \varepsilon > \frac{23}{5}. \quad (\text{FWP19})$$

For the other constant, $H_n - \log n$ decreases strictly to γ_E . Strict decrease follows from $\log(1 + 1/n) > 1/(n + 1)$, obtained by integrating $1/(1 + t)$ on $[0, 1/n]$. Hence $\gamma_E < H_{10} - \log 10$. The elementary bound $\pi < 22/7$ can be verified without a decimal approximation by the strictly positive integral

$$\begin{aligned} 0 &< \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx \\ &= \int_0^1 \left(x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2} \right) dx = \frac{22}{7} - \pi. \end{aligned} \quad (\text{FWP20})$$

The equality of integrands is polynomial multiplication; the last integral uses $\arctan 1 = \pi/4$. Using additionally $\log(1 + t) < t$ for $t > 0$, obtained by integrating $(1 + t)^{-1} < 1$, yields

$$\begin{aligned} c = \log(4\pi) + \gamma_E &< H_{10} + \log(2\pi/5) < H_{10} + \log(44/35) \\ &< H_{10} + 9/35 = \frac{7381}{2520} + \frac{9}{35} = \frac{8029}{2520}. \end{aligned} \quad (\text{FWP21})$$

Every inequality used in this constant is strict.

2.5 The full form and the exact remaining sign

On the two-moment test space FWP3, the complete Weil form in the fixed convention is $Q(g) = -W_{\text{arch}}(f) - \sum_p W_p(f)$. Thus FWP9 and FWP17 imply

$$Q(g) > N \left[3 \log \coth \varepsilon - 3c - 27\varepsilon - 6\varepsilon^2 - 6/7 - 3 \right]. \quad (\text{FWP22})$$

FWP18 gives $27\varepsilon + 6\varepsilon^2 < 27/100 + 6/10000 = 1353/5000$. Substitute this, FWP19, and FWP21, keeping the exact rational values:

$$\begin{aligned} \frac{Q(g)}{N} &> \frac{69}{5} - 3 \frac{8029}{2520} - \frac{1353}{5000} - \frac{6}{7} - 3 \\ &= \frac{5981}{52500} = \frac{1}{10} + \frac{731}{52500} > \frac{1}{10}. \end{aligned} \quad (\text{FWP23})$$

This proves FWP1 for every pair of distinct primes and all four sign choices, without altering the original bump, its derivatives, its width, its norm, or the Haar measures.

The intermediate Archimedean bound is also retained explicitly. Put $\alpha_u = 2N(\log u)/\sqrt{u}$ and $\alpha_v = 2N(\log v)/\sqrt{v}$. FWP17–FWP21 give, before subtracting any finite-prime contribution,

$$-W_{\text{arch}}(f) > \left(3 + \frac{5981}{52500} \right) N > \alpha_u + \alpha_v + \frac{N}{10}. \quad (\text{FWP23a})$$

The last inequality uses $\alpha_u + \alpha_v \leq 8N/e < 3N$. This holds for all four sign choices in FWP2 and keeps the exact Archimedean value attached to the actual chosen test.

For the sign choice $\sigma = 1, \tau = -1$, let $\alpha_u = 2N(\log u)/\sqrt{u}$ and $\alpha_v = 2N(\log v)/\sqrt{v}$. The original support vector $\mathbf{h} = \delta_{\emptyset} - \delta_{\{u\}} - \delta_{\{v\}} + \delta_{\{u,v\}}$ has joint observation $(W_u + W_v)^{\otimes r} - W_u^{\otimes r} - W_v^{\otimes r}$, by expanding its four support states. FWP8 consequently gives, on the very same $f = g * g^*$ used in the positive full-form bound,

$$\mathcal{V}_{2m}(\mathbf{h})(f^{\otimes 2m}) = (\alpha_u - \alpha_v)^{2m} - \alpha_u^{2m} - \alpha_v^{2m} < 0 \quad (m \geq 1), \quad Q(g) > N/10. \quad (\text{FWP24})$$

The strict negative sign follows from $|\alpha_u - \alpha_v| < \max(\alpha_u, \alpha_v)$. This proves exactly how the negative mixed observation coexists with a strictly positive complete Weil form. These explicit tests are excluded as full-Weil counterexamples by the calculated bound; their mixed distribution and its nonzero evaluations remain intact.

3 The common Archimedean term in the complete mixed observation

Retain the test $g = g_{u,v;1,-1}$, its convolution square $f = g * g^*$, the mass $N = \int z(t)^2 dt$, and the exact positive numbers $\alpha_u = 2N(\log u)u^{-1/2}$ and $\alpha_v = 2N(\log v)v^{-1/2}$ of PME21–PME26. These symbols have their test-function meaning in this section. In particular N here is not the companion source cutoff used in SBT. The complete Archimedean number is

$$b = -W_{\text{arch}}(f), \quad b > \alpha_u + \alpha_v + \frac{N}{10}. \quad (\text{FWJ1})$$

The strict inequality is the intermediate estimate in FWP: it bounds b below by $3N + (5981/52500)N$, whereas $\alpha_u + \alpha_v < 3N$. Thus (FWJ1) follows from the calculated Archimedean integral, not an assigned positive parameter.

3.1 The exact four-state distribution map

Use the same support space $V = \mathbb{C}[\mathcal{P}(\{u, v\})]$ and original signed state $\mathbf{h} = \delta_\emptyset - \delta_{\{u\}} - \delta_{\{v\}} + \delta_{\{u, v\}}$. For $p = u, v$, put $E_p \delta_A = \mathbf{1}_{p \in A} \delta_A$, and let $D_{\mathbf{h}}$ have diagonal $(1, -1, -1, 1)$ in that order. The full common-Archimedean receiving operator on this finite support space is exactly

$$Q(f) = -W_{\text{arch}}(f)I - W_u(f)E_u - W_v(f)E_v = bI - \alpha_u E_u + \alpha_v E_v. \quad (\text{FWJ2})$$

All finite places outside u, v have already been proved to vanish for this test in PME25. Its two pole moments vanish by PME22. Consequently the diagonal entry at $A = \{u, v\}$ is the actual complete Weil value of this test, with neither an omitted finite prime nor an omitted Archimedean or pole term. At a proper subset A , the entry is the exact corresponding support-weighted face: the same Archimedean functional, with the indicated finite-place weights. Explicitly,

$$\begin{array}{c|cccc} A & \emptyset & \{u\} & \{v\} & \{u, v\} \\ \hline Q_A(f) & b & b - \alpha_u & b + \alpha_v & b - \alpha_u + \alpha_v \end{array} \quad Q_A(f) > N/10 \quad \text{for every } A. \quad (\text{FWJ3})$$

The inequality follows by subtracting at most α_u in (FWJ1); adding α_v cannot decrease a value. This proves positivity of all four displayed values directly.

For completeness, the linear distribution map prior to this evaluation is $Q\delta_A = -W_{\text{arch}} - \sum_{p \in A} W_p$. Define its joint map with the original support diagonal, not by taking a tensor power after applying it to $\mathbf{h}$:

$$\Delta_r \delta_A = \delta_A^{\otimes r}, \quad \mathcal{Q}_r = \mathcal{Q}^{\otimes r} \Delta_r, \quad \mathcal{Q}_r(\mathbf{h})(f^{\otimes r}) = \text{Tr}_V[D_{\mathbf{h}} Q(f)^r]. \quad (\text{FWJ4})$$

The last equality follows by evaluating each of the four basis vectors: both sides are $\sum_A h_A Q_A(f)^r$. Thus the shared Archimedean term has been transported through the same full four-state diagonal used in PME15 and PME29. No infinite operator trace is involved in (FWJ4).

3.2 Its sign, all tensor orders, and exact surviving rate

Substituting (FWJ3) gives the complete expression

$$J_r := \mathcal{Q}_r(\mathbf{h})(f^{\otimes r}) = b^r - (b - \alpha_u)^r - (b + \alpha_v)^r + (b - \alpha_u + \alpha_v)^r. \quad (\text{FWJ5})$$

It is zero for $r = 1$. For every integer $r \geq 2$, applying the fundamental theorem of calculus first to x , then to y , yields

$$J_r = -r(r-1) \int_0^{\alpha_u} \int_0^{\alpha_v} (b - x + y)^{r-2} dy dx < 0. \quad (\text{FWJ6})$$

To verify the sign in the identity, set $F(x, y) = (b - x + y)^r$. Then $\partial_x \partial_y F = -r(r-1)(b - x + y)^{r-2}$, and the boundary value of its integral is $F(\alpha_u, \alpha_v) - F(\alpha_u, 0) - F(0, \alpha_v) + F(0, 0)$, exactly (FWJ5). By (FWJ1), $b - x + y \geq b - \alpha_u > N/10 > 0$ on the entire rectangle. Its side lengths are strictly positive, so the inequality in (FWJ6) is strict, also when r is odd. For $r = 2$ the integral is constant and gives

$$J_2 = -2\alpha_u \alpha_v = -8N^2(\log u)(\log v)(uv)^{-1/2}. \quad (\text{FWJ7})$$

The common Archimedean contribution cancels from this second difference, though it has remained in all four face values and their positive lower bounds.

The exact asymptotic rate is

$$\lim_{r \rightarrow \infty} (-J_r)^{1/r} = b + \alpha_v > 0. \quad (\text{FWJ8})$$

Indeed divide $-J_r$ by $(b + \alpha_v)^r$ in (FWJ5). The three other bases b , $b - \alpha_u$, and $b - \alpha_u + \alpha_v$ are strictly positive and strictly smaller than $b + \alpha_v$, so their ratios raised to r tend to zero. The quotient tends to one and is positive by (FWJ6). Its logarithm divided by r tends to zero, which proves (FWJ8). All bases, including the exact Archimedean value b , have been retained.

The calculated relationship. The single full Weil form $Q(g) = Q_{\{u,v\}}(f)$ is positive with the explicit lower bound in FWP. The same test, carried through the full shared-Archimedean support diagonal, has strictly negative joint trace at every order at least two. The exact connecting map is (FWJ4), and the exact negative quantity is (FWJ6). Thus positivity of all one-copy face values does not imply positivity of this signed mixed-support trace. This is a constructed counterexample to that particular implication, not a zero off the critical line and not a negative value of the complete one-variable Weil form. It also shows explicitly that inserting the common Archimedean term alone does not make this mixed quantity disappear under tensor powers.

4 The original quartet inside the relative-prime distribution space

The input is the original companion object in TDP1–4 of the retained reconstruction, together with the relative-prime operator in (PD11), (PD20)–(PD21) of the accompanying 031 source. We construct an injective intertwining map, calculate its exact finite boundaries, and determine the growth space occupied by every original vector. This calculation does not assert that this distribution space is the image of compactly supported cohomology in ordinary cohomology. It gives the concrete distribution object and the precise boundary that a comparison with that image must calculate. The measure and every original spectral parameter are kept throughout.

4.1 Original coordinates and the actual local tensor operator

Keep $k \geq 17$, $k \equiv 1 \pmod{4}$, $q = (k+1)^2$, $\delta, \gamma > 0$, and each original source cutoff $q-1 \leq N \leq 2q$. Write

$$\lambda_{ab} = (2b-k)\gamma - i(2a-k)\delta, \quad Q_k(y) = \prod_{a,b=0}^k (y - \lambda_{ab}), \quad (\text{SBT1})$$

$$E = \mathbb{C}[y]/(Q_k), \quad M[f] = [yf], \quad F_p = e^{i(\log p)M}, \quad \tilde{F}_p = p^{k/2}F_p.$$

Here p is any rational prime. Let $\langle v, w \rangle_N = v^* G_N w$ be the complete original positive source metric. No diagonalization of that metric is made. All identities below hold for each such G_N ; none uses an estimate or substitution for any of its entries. In particular they apply separately at all four original cutoffs $q-1, q, 2q-1, 2q$, with their metrics unchanged.

The roots are distinct, because equality of real and imaginary parts forces both indices to agree. Thus

$$\ell_{ab}(y) = \prod_{(c,d) \neq (a,b)} \frac{y - \lambda_{cd}}{\lambda_{ab} - \lambda_{cd}}, \quad P_{ab}v = v(\lambda_{ab})\ell_{ab}, \quad \omega_{ab} = p^{(2a-k)\delta} e^{i(2b-k)\gamma \log p} \quad (\text{SBT2})$$

are defined without choosing branches of a logarithm. Evaluation proves $\sum P_{ab} = I$, $P_{ab}P_{cd} = 0$ for different indices, $M\ell_{ab} = \lambda_{ab}\ell_{ab}$, and $F_p\ell_{ab} = \omega_{ab}\ell_{ab}$. The projections refer to M , so coincident phases of the F_p eigenvalues do not identify labels.

On $L^2(\mathbb{Q}_p, dx)$, where $\mu(\mathbb{Z}_p) = 1$, the actual relative-prime dilation is

$$(\mathcal{U}_p F)(x) = \sqrt{p}F(x/p), \quad T_p^{\text{rel}} = \sqrt{p}\mathcal{U}_p. \quad (\text{SBT3})$$

It comes from the p -to-one multiplication map on $\mathbb{Q}_p/\mathbb{Z}_p$ in (RP30a), (PD7)–(PD11), rather than an assigned weight on a support point. Indeed a change of variables gives $\int p|F(x/p)|^2 dx = \int |F(y)|^2 dy$, since $|p|_p = p^{-1}$. Its inverse is $p^{-1/2}F(px)$, proving unitarity directly. Use its actual k th tensor product

$$\mathcal{U}_{p,k} = \mathcal{U}_p^{\otimes k}, \quad \mathcal{T}_{p,k} = (T_p^{\text{rel}})^{\otimes k} = p^{k/2}\mathcal{U}_{p,k} \quad \text{on } L^2(\mathbb{Q}_p^k, d\mathbf{x}). \quad (\text{SBT4})$$

The factor $p^{k/2}$ is therefore the original uncentred factor in (SBT1), with exactly k local tensor factors.

For every $n \in \mathbb{Z}$ define the following actual functions:

$$h_n(\mathbf{x}) = p^{kn/2}(1 - p^{-1})^{-k/2} \mathbf{1}_{(p^n \mathbb{Z}_p^\times)^k}(\mathbf{x}). \quad (\text{SBT5})$$

The set in (SBT5) has measure $p^{-kn}(1 - p^{-1})^k$. The h_n are consequently orthonormal, and substitution in (SBT3) gives $\mathcal{U}_{p,k} h_n = h_{n+1}$. Their closed span $\mathcal{R}_{p,k}$ is a reducing subspace, because the inverse carries h_n to h_{n-1} . It is the indicated diagonal-annulus channel, not all of $L^2(\mathbb{Q}_p^k)$. The coefficient map

$$\ell^2(\mathbb{Z}; E, G_N) \longrightarrow \mathcal{R}_{p,k} \hat{\otimes} E, \quad (x_n) \longmapsto \sum_{n \in \mathbb{Z}} h_n \otimes x_n \quad (\text{SBT6})$$

is an isometry onto: orthogonality proves the norm equality first for finite sums, and completeness proves the assertion for square-summable sequences. In these exact coordinates, $\mathcal{U}_{p,k} \otimes I_E$ is $(Sx)_n = x_{n-1}$.

4.2 A complete distribution object and its inverse evaluation

Let $\mathcal{D}_{p,k}(E) = E^{\mathbb{Z}}$. We use the word distribution here for the full linear dual of the finite-support test space $c_{00}(\mathbb{Z}; E^*)$: the pairing is $\langle x, \varphi \rangle = \sum_n \varphi_n(x_n)$. Thus every pairing is finite; no analytic continuation or unproved convergence is used. The shift S is defined on this whole product space. Define

$$\mathcal{J}_p : E \longrightarrow \mathcal{D}_{p,k}(E), \quad (\mathcal{J}_p v)_n = F_p^{-n} v = \sum_{a,b=0}^k \omega_{ab}^{-n} P_{ab} v. \quad (\text{SBT7})$$

Then the following identities hold exactly:

$$\text{ev}_0 \mathcal{J}_p = I_E, \quad S \mathcal{J}_p = \mathcal{J}_p F_p, \quad p^{k/2} S \mathcal{J}_p = \mathcal{J}_p \tilde{F}_p. \quad (\text{SBT8})$$

For the second identity, the n th term on the left is $F_p^{-(n-1)} v = F_p^{-n} F_p v$; the other identities are immediate substitutions. In particular the map is injective. Its image is the fully specified subspace

$$\mathcal{C}_p(E) = \{x \in E^{\mathbb{Z}} : x_{n-1} = F_p x_n \text{ for every } n \in \mathbb{Z}\}. \quad (\text{SBT9})$$

To prove equality of the image and this space, apply the recurrence in both directions, using invertibility of F_p , to get $x_n = F_p^{-n} x_0$. Evaluation at zero is its inverse. Thus the counterfactual companion object has been constructed as one coherent distribution subspace, with its original q labels, rather than just a collection of spectral inequalities.

Its original metric is recovered exactly by

$$\langle x, x' \rangle_{\mathcal{C},N} = \langle x_0, x'_0 \rangle_N, \quad \langle \mathcal{J}_p v, \mathcal{J}_p w \rangle_{\mathcal{C},N} = v^* G_N w. \quad (\text{SBT10})$$

This positive metric is a metric on $\mathcal{C}_p(E)$; it is not the generally divergent sum of the annulus Hilbert norms. In this retained metric the precise local weight defect is

$$\|\mathcal{J}_p \tilde{F}_p v\|_{\mathcal{C},N}^2 - p^k \|\mathcal{J}_p v\|_{\mathcal{C},N}^2 = p^k v^* (F_p^* G_N F_p - G_N) v. \quad (\text{SBT11})$$

For $v = \ell_{ab}$ it is $p^k(p^{2(2a-k)\delta} - 1) \|\ell_{ab}\|_N^2$. Both signs occur among the original labels. This follows by inserting their exact eigenvalues, not by replacing the metric with the cardinal basis metric.

4.3 The precise growth space of the original object

Define $s(\mathbb{Z}; E^*)$ to consist of sequences for which all seminorms $\sup_n (1 + |n|)^j \|\varphi_n\|_{N,*}$ are finite. A sequence $x \in E^{\mathbb{Z}}$ extends to a continuous functional on this space precisely when

$$\|x_n\|_N \leq C(1 + |n|)^m \quad (n \in \mathbb{Z}) \quad (\text{SBT12})$$

for some C, m . Here is a proof of both directions. This bound makes the pairing absolutely convergent, bounded by the $(m+2)$ nd seminorm times $C \sum_n (1+|n|)^{-2}$. Conversely, continuity at zero bounds a linear functional by a constant times finitely many of the defining seminorms. Increasing their largest index dominates all of them. Testing on one index n with a norm-one covector attaining $\|x_n\|_N$ proves (SBT12). Finite-support sequences are dense in s by truncation and the bound of each tail using the next seminorm, so the continuous extension, when it exists, is unique. Call this space of polynomial-growth distributions $s'(\mathbb{Z}; E)$.

For a nonzero original v put

$$R_p(v) = \delta(\log p) \max\{|2a - k| : P_{ab}v \neq 0\}. \quad (\text{SBT13})$$

Then

$$\lim_{H \rightarrow \infty} \frac{1}{H} \log \max\{\|F_p^H v\|_N, \|F_p^{-H} v\|_N\} = R_p(v). \quad (\text{SBT14})$$

The upper bound follows from the finite sum (SBT7) and the triangle inequality, with the constant $\sum_{a,b} \|P_{ab}v\|_N$. For the lower bound choose a nonzero $P_{ab}v$ attaining the maximum. Use F_p^H if $2a - k > 0$, and F_p^{-H} if $2a - k < 0$. Since P_{ab} commutes with F_p , $\|F_p^{\pm H} v\|_N \geq \|P_{ab}\|_{N \rightarrow N}^{-1} e^{HR_p(v)} \|P_{ab}v\|_N$. Taking logarithms and dividing by H proves (SBT14); all constants are independent of H , and no orthogonality of the P_{ab} was used. Because k is odd, every $|2a - k| \geq 1$. Consequently

$$\mathcal{C}_p(E) \cap s'(\mathbb{Z}; E) = \{0\}, \quad R_p(v) \geq \delta \log p \quad (v \neq 0). \quad (\text{SBT15})$$

This is a calculated obstruction in this specific annulus comparison. It does not identify the global arithmetic cohomology with s' . For contrast within the same distribution space, the rank-one sequence $x_n = \zeta^{-n}w$ with $|\zeta| = 1$ and $w \neq 0$ is bounded, is a generalized eigenvector of S , and lies in s' but not in ℓ^2 . Thus passing to distributions here does retain the entire unit-circle spectral boundary; (SBT15) is not the assertion that a bilateral shift has no Hilbert eigenvectors.

4.4 Finite boundaries, with all original phases retained

Let $H \geq 0$ be an integer annulus cutoff, distinct from the original source cutoff N . Define an actual finite Hilbert vector

$$\mathcal{J}_{p,H}v = \sum_{n=-H}^H h_n \otimes F_p^{-n}v. \quad (\text{SBT16})$$

Its norm, with $v_{ab} = v(\lambda_{ab})$ and $G_{ab,cd}^{\text{card}} = \ell_{ab}^* G_N \ell_{cd}$, is exactly

$$\|\mathcal{J}_{p,H}v\|^2 = \sum_{n=-H}^H v^*(F_p^{-n})^* G_N F_p^{-n}v = \sum_{ab,cd} \overline{v_{ab}} v_{cd} G_{ab,cd}^{\text{card}} \sum_{n=-H}^H (\overline{\omega_{ab}} \omega_{cd})^{-n}. \quad (\text{SBT17})$$

The finite geometric sum is intentionally retained also when its argument is one; it then equals $2H+1$. This expression preserves every phase and every off-diagonal entry of the original metric. The intertwining defect has exactly two boundary terms:

$$\begin{aligned} B_{p,H}v &= (\mathcal{J}_{p,k} \otimes I_E) \mathcal{J}_{p,H}v - \mathcal{J}_{p,H} \tilde{F}_p v \\ &= p^{k/2} (h_{H+1} \otimes F_p^{-H}v - h_{-H} \otimes F_p^{H+1}v), \\ \|B_{p,H}v\|^2 &= p^k (\|F_p^{-H}v\|_N^2 + \|F_p^{H+1}v\|_N^2). \end{aligned} \quad (\text{SBT18})$$

Indeed shifting the first finite sum changes its range to $-H+1, \dots, H+1$. At every common index the coefficient is $p^{k/2} F_p^{1-n}v$ in both sums. The two remaining indices give the displayed terms, and their orthogonality proves the norm identity. In particular $B_{p,H}$ is injective and its positive quadratic form has no omitted contribution from the original observation kernel.

For any original cardinal vector put $r_{ab} = |\omega_{ab}| = p^{(2a-k)\delta}$. Substitution gives

$$\frac{\|B_{p,H}\ell_{ab}\|^2}{\|\mathcal{J}_{p,H}\ell_{ab}\|^2} = p^k \frac{r_{ab}^{-2H} + r_{ab}^{2H+2}}{\sum_{n=-H}^H r_{ab}^{-2n}} \rightarrow p^k |r_{ab}^2 - 1| > 0. \quad (\text{SBT19})$$

For $r > 1$, divide numerator and denominator by r^{2H} ; the denominator tends to $1/(1 - r^{-2})$, and the numerator to r^2 . For $r < 1$, divide by r^{-2H} instead, obtaining denominator $1/(1 - r^2)$ and numerator one. These calculations prove the limit and its sign for every original label. At $r = 1$ the same finite expression is $2p^k/(2H + 1)$ and tends to zero; this last identity is the separate unit-circle test described after (SBT15).

For any unit vector z in $L^2(\mathbb{Q}_p^k) \hat{\otimes} (E, G_N)$ and any complex number λ , the reverse triangle inequality gives

$$\|((\mathcal{J}_{p,k} \otimes I_E) - \lambda I)z\| \geq ||\lambda| - p^{k/2}|. \quad (\text{SBT20})$$

It uses the exact equality $\|\mathcal{J}_{p,k}z\| = p^{k/2}$. Therefore no alternative sequence of unit Hilbert vectors can make the eigenvalue residual tend to zero at an original off-circle $\lambda = p^{k/2}\omega_{ab}$. Formula (SBT19) is the exact boundary of the constructed map, and (SBT20) also proves the obstruction for all such Hilbert approximations. Neither statement erases the distribution vector (SBT7), which exists and has the exact growth calculated above.

4.5 The complete lattice and the original observation kernel

Let L be the original bounded distributive support lattice and $G_L(\mathbb{C}) = \{(0, \lambda) : \lambda \in L\} \cup \{(c, 1_L) : c \in \mathbb{C}\}$, with join in addition and meet in multiplication. For a complex vector space V retain its synchronized carrier $V^\uparrow = \{(0, \lambda)\} \cup \{(v, 1_L)\}$. Addition is $(v, \lambda) + (w, \mu) = (v + w, \lambda \vee \mu)$; scalar multiplication is $(c, \sigma)(v, \lambda) = (cv, \sigma \wedge \lambda)$. For any complex-linear map A , its supported lift is $A^\uparrow(v, \lambda) = (Av, \lambda)$. It preserves both operations by substitution, including when a nonzero amplitude cancels. Thus

$$\mathcal{J}_p^\uparrow : E^\uparrow \hookrightarrow \mathcal{D}_{p,k}(E)^\uparrow, \quad (v, \lambda) \mapsto (\mathcal{J}_p v, \lambda) \quad (\text{SBT21})$$

is injective, preserves each labelled zero, and satisfies the lifted identities (SBT8). In particular the inverse evaluation recovers $(0, \lambda)$, not just its vanishing complex amplitude. The lifted metric is $\langle (v, \lambda), (w, \mu) \rangle^\uparrow = (v^* G_N w, \lambda \wedge \mu)$ and obeys (SBT10) exactly. These are the full- L carriers proved to receive the actual companion path closure in CLM8–17; no invertible free-semimodule lift is asserted.

Finally retain $J : B \rightarrow E$, $K = \ker \Lambda$, $P_B = JJ^\dagger$, $Q = I - P_B$ and the isometric inclusion $\iota_K : K \rightarrow E$ of TDP4. The columns

$$\mathcal{J}_p J : B \rightarrow \mathcal{C}_p(E), \quad \mathcal{J}_p \iota_K : K \rightarrow \mathcal{C}_p(E), \quad x = \mathcal{J}_p J(J^\dagger x_0) + \mathcal{J}_p(Qx_0) \quad (\text{SBT22})$$

give the complete original decomposition inside the new object. The last formula follows from $x = \mathcal{J}_p x_0$ and $JJ^\dagger + Q = I$. Both summands and their common ambient metric are retained. The failure of the first column alone to intertwine its compressed operator is exactly

$$p^{k/2} S \mathcal{J}_p J - \mathcal{J}_p J(J^\dagger \tilde{F}_p J) = \mathcal{J}_p Q \tilde{F}_p J. \quad (\text{SBT23})$$

Use (SBT8) on the left and factor $I - JJ^\dagger = Q$ to prove the identity. Thus the full original kernel return has an explicit image in the distribution comparison; it is not discarded by the passage to the local tensor space.

Consequence for the purity calculation. The degree- p relative operator has an exact weight- k tensor norm, while the original off-critical companion space embeds into its explicit distribution channel with exponential growth at least $\delta \log p$. Its finite comparison has the positive boundary

ratio (SBT19). This determines both the connecting map and the surviving object. It is not a contradiction within the definitions: the distribution space admits these vectors, and its evaluation metric is explicitly (SBT10). No identification of this space with the global interior cohomology or its trace is asserted. That identification cannot be supplied merely by citing the local norm identity, because the exact transported object and its nonvanishing boundary have now been calculated.

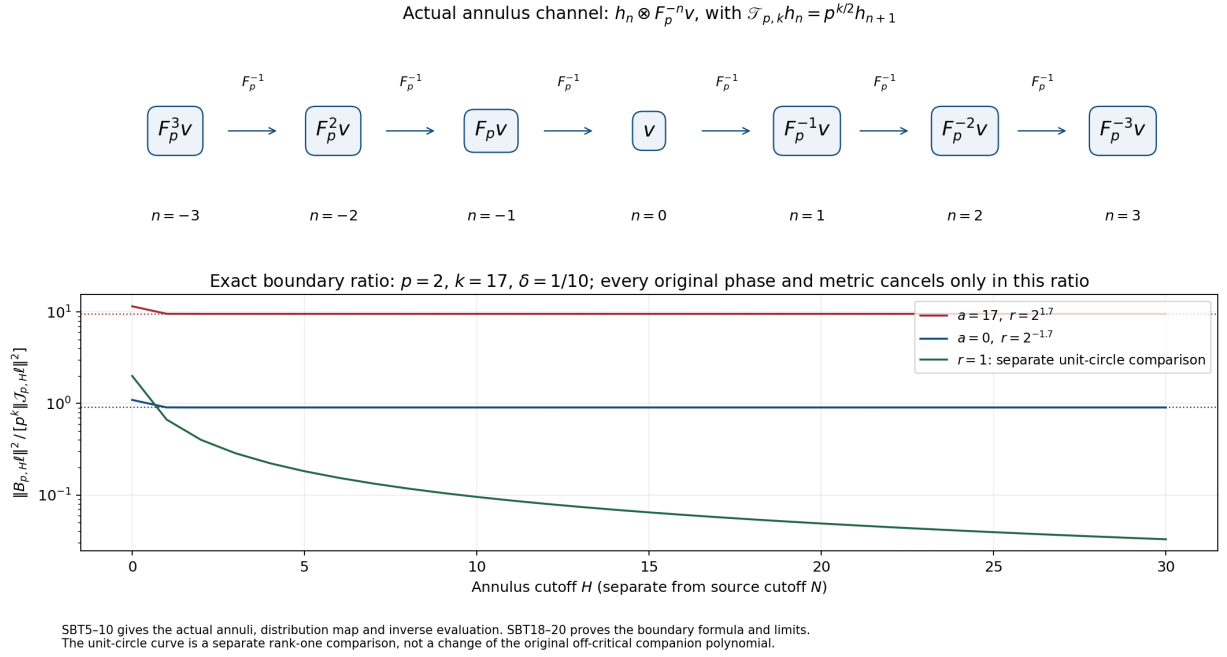


Figure 2: SBT5–10 constructs the actual annulus channel and its original E -valued coefficient sequence. The arrows between coefficients are F_p^{-1} ; the spatial shift itself acts on h_n . SBT18–20 proves the displayed boundary ratio and its nonzero off-circle limits. The unit-circle curve is the separate comparison explicitly defined after SBT15, not a modification of the original off-critical polynomial. The inherited relative degree operator is proved in PME1–5 and the supplied 031 PD1–21 source.